Ultimate Nutrition®

Clean Whey Protein

Protein is probably the most important supplement for all athletes, but it is especially indispensable for those involved in long-term, high-intensity endurance training. It is the fuel for the build-up, mass and toning of the muscles. In fact, protein is essential for growth and development. The proteins that make up the human body are not obtained directly from the diet; rather, dietary protein is broken down into its constituent parts known as amino acids, which are the basic building blocks of life. That means that it is the amino acids, and not protein per se, which are essential nutrients. Of the 20 amino acids that the body needs to make protein, eleven are designated as nonessential amino acids, because they can be produced by the human body from other amino acids and do not need to be obtained from the diet. The remaining nine amino acids cannot be synthesized by the body and are, therefore, called essential amino acids. These amino acids must be obtained by degradation of the dietary protein. All of these essential amino acids must be present in the body in order for it to build and repair muscle. The manner in which the body uses protein during and after exercise is quite different from the way it utilizes carbohydrates or fat for energy. In the hours after exercise, the body begins to build the so-called structural proteins from amino acids for muscle repair and growth, and remodeling the tissues needed for performance. This is one of the reasons that athletes should pay particular heed to nutrition for complete recovery, and make it an integral part of their workout regimen. Exercise duration and intensity are important factors in determining which of the nutrient fuels are burned to gain energy. An exercise regimen comprising of low-to-moderate intensity of long duration necessitates large quantities of fuel--oftentimes more than carbohydrates and fat reserves in the body can provide. Therefore, the body begins to utilize structural and functional proteins during long-term aerobic exercise. In contrast, high-intensity exercise of short duration mostly uses glucose, which spares the protein reservoir of the body. To meet the needs of athletes, researchers at Ultimate Nutrition have developed Ultimate Nutrition Clean Whey to supply all of the essential and non-essential amino acids to build muscle after intense exercise of both short and long duration. It is a customized whey, isolated by a complex low temperature processing system, that utilizes proprietary micro- and ultra-filtration to ensure the highest-quality whey protein.

Ultimate Nutrition Clean Whey contains all the whey protein fractions such as Beta Lactoglobulin, Alpha Lactalbumin, Glycomacropeptide, Immunoglobulins, Proteose Peptones, Serum Albumin, Lactoferrin and Lactoperoxidase.

Since exercise can take a toll on the immune system, Ultimate Nutrition Clean Whey is the nutritional protein shake of choice for those who lead an active lifestyle. Ultimate Nutrition Clean Whey supports muscle maintenance, buildup and toning for anyone from weekend warriors to hardcore bodybuilders. Furthermore, it provides wholesome support for compromised or wasting muscle due to the aging process. Ultimate Nutrition Clean Whey is the nutrition for champions, no matter what your goals are!

SELECTED REFERENCES

1) Chang, T., Wen, A. and Goldberg, A., "The Metabolic fates of Amino Acids and the Formation of Glutamine in Skeletal Muscle," J. Biol. Chem.: 253, 3685, 1978 2) DiPrampero, P., "Energetics of Muscular Exercise," Biochem. Pharmacol.: 89, 143, 1981 3) Jacobs, I., "Sprint Training Effects on Muscle Myoglobin, Enzymes, Fiber Types, and Blood Lactate," Med. Sci. Sports Exer.: 19, 369, 1987 4)Murray, M. and Pizzorno, J., Encyclopedia of Natural Medicine, Prima Publishing, Rocklin, California, 1998 5)Champe, P. and Harvey, R., Biochemistry, Lippincott, Williams and Wilkins, Philadelphia, Pennsylvania, 1994 6)Holloszy, J., "Adaptation of Skeletal Muscle to Endurance Exercise," Med Sci Sports: 7, 155, 1975

FREQUENTLY ASKED QUESTIONS

1) How is Clean Whey taken?

It can be ingested in many different ways. Most commonly, it is either mixed with water or a nutritious beverage. It can be taken numerous times during the day to boost energy, but it is most beneficial if taken immediately after intense exercise. Two or three servings daily are recommended for those involved in vigorous exercise to ensure maintenance of positive nitrogen balance. This allows the body to repair and build muscle and minimize muscle breakdown.

2) How Does Clean Whey Boost Immunity?
Whey protein is quite singular in its property to boost immunity by increasing the body's own antioxidants. In addition, Clean Whey Whey is blended such that the immune boosting proteins are retained, which include b-globulin, a-lactalbumin, glycomacropeptide, immunoglobulins and, among others, lactoferrin. All constituents in this category inherently boost the immune system, which does become weakened somewhat with sustained exercise regimens.

3) How Does Clean Whey Whey Taste?
Clean Whey has been blended with delicious natural flavors and sweetened with Stevia to taste great, overcoming the chemical taste that the amino acids can have.

4) Does Clean Whey Induce Growth Hormone?
Insulin growth factor-1 (IGF-1) is the hormone released during growth hormone metabolism. The amount of IGF-1 produced is determines the extent of growth hormone on growth in general. Studies have shown that IGF-1 rises in direct proportion to the quantity and quality of protein in the diet. Hydrolyzing whey increases its biological value (please see Ultimate Nutrition's Webpage on Super Amino 2000) that, in turn, increases the release of IGF-1 making whey the best body-building protein available by far.